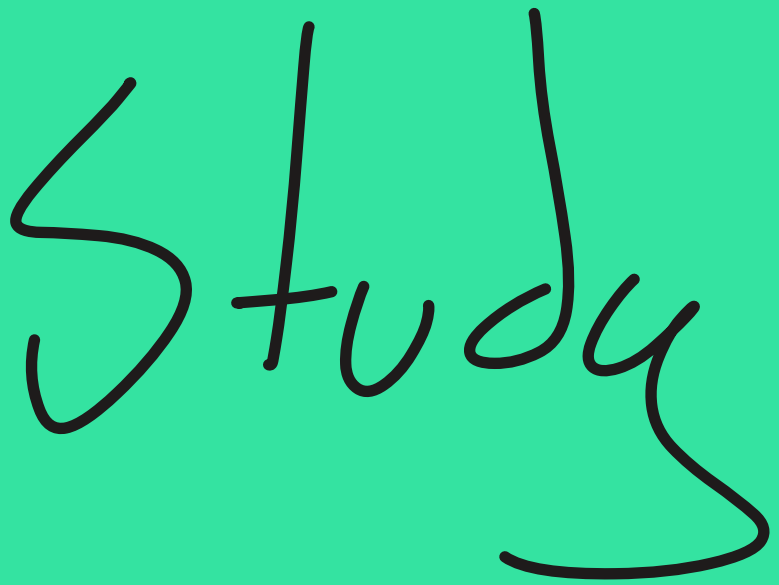

Study

Simplify

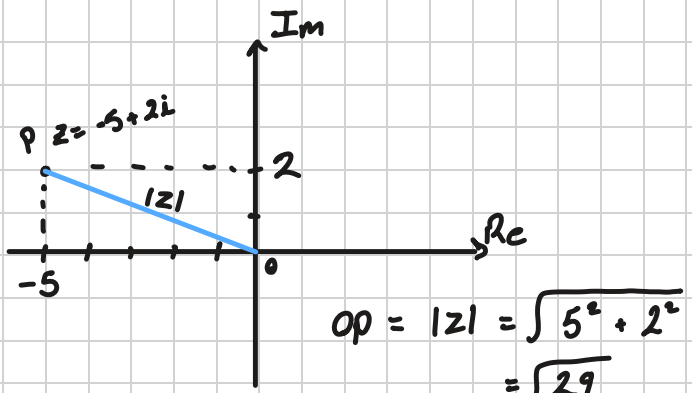
a) $\frac{2+3i}{4-i} \times \frac{4+i}{4+i} = \frac{(2+3i)(4+i)}{(4+i)(4-i)} = \frac{8-3+(2+12)i}{16+1} = \frac{5}{17} + \frac{14}{17}i$

b) $\frac{i}{1+i\sqrt{3}} = \frac{i(1-i\sqrt{3})}{(1+i\sqrt{3})(1-i\sqrt{3})} = \frac{\sqrt{3}+i}{1+3} = \frac{\sqrt{3}}{4} + \frac{1}{4}i$

① $z = -5 + 2i$

$Re(z) = -5$

$Im(z) = 2$



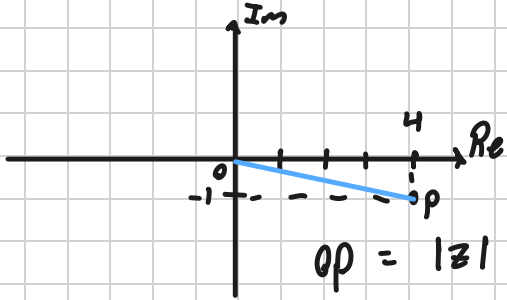
$OP = |z| = \sqrt{5^2 + 2^2}$

$= \sqrt{29}$

② $z = 4 - i$

$Re(z) = 4$

$Im(z) = -1$



$OP = |z| = \sqrt{4^2 + 1}$

$= \sqrt{17}$

$y = f(x)$ differentiable

$y' =$

Complex numbers

① $z = -5 + 2i$

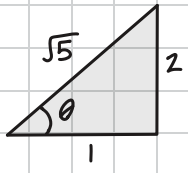
real: -5 imaginary: $2i$

② $z = 1 + 2i$

$r = |z| = \sqrt{a^2 + b^2}$

$= \sqrt{1^2 + 2^2}$

$= \sqrt{5}$



$\theta = \tan^{-1}\left(\frac{2}{1}\right)$

$= 63^\circ$

$z = a + bi = r\left(\frac{a}{r} + \frac{b}{r}i\right)$

$= r(\cos\theta + i\sin\theta)$ polar form

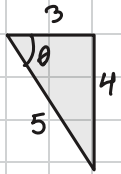
$= \sqrt{5}(\cos 63^\circ + i\sin 63^\circ)$

(3)

$z = 3 - 4i$

$r = \sqrt{3^2 + 4^2}$

$= 5$



$\tan^{-1}\left(\frac{-4}{5}\right) = -39° \leftarrow$ degrees

$\frac{-4}{5} = -0.8 \leftarrow$ radians

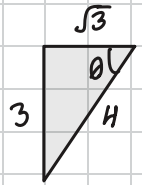
(4)

$z = -\sqrt{3} - 3i$

$r = \sqrt{3 + 9}$

$= \sqrt{12}$

$= 4$



$\theta = \frac{3}{\sqrt{3}}$ rad

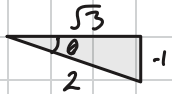
$\theta = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right)$

(5)

$z = \sqrt{3} - i$

$r = \sqrt{3 + 1}$

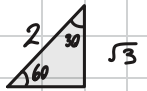
$= 2$



$\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \theta$

$\theta = -\frac{1}{\sqrt{3}}$

$= -\frac{\pi}{6}$ rads



$\tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$

30°

(6)

$z = 3i$

$r = \sqrt{0 + 3^2}$

$= 3$

YAY

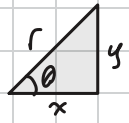


$\theta = \frac{\pi}{2}$

(7) $|z| = 5$, $\arg(z) = \tan^{-1}\frac{3}{4}$

$z = 4 + 3i$

(8) $|z| = \frac{1}{2}$, $\arg(z) = -\frac{\pi}{3}$



$\cos\theta = \frac{x}{r}$

$x = r\cos\theta$

$x = r\cos\theta$
$y = r\sin\theta$ ⟶ $x = \frac{1}{2}\cos\left(-\frac{\pi}{3}\right) = \frac{1}{4}$

$y = \frac{1}{2}\sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{4}$

$z = \frac{1}{4} - \frac{\sqrt{3}}{4}i$

Complex conjugate

(9) $-3 - 5i$

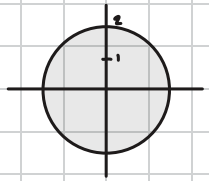
$\overline{z} = -3 + 5i$

(10) $z = 2 - i$

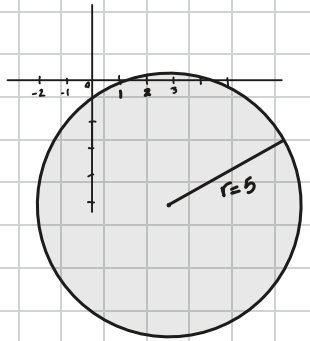
$\overline{z} = 2 + i$

(11) Sketch to satisfy

$|z| \leq 2$



(12) $|z - 3 + 4i| \leq 5$



(13) Simplify

$i - (3 - 2i) + (7 - 3i)$

$= i - 3 + 2i + 7 - 3i$

Re: $-3 + 7 = 4$

Im: $1i + 2i - 3i = 0i$

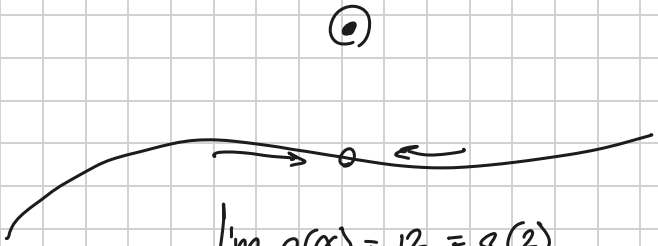
$= 4$

(14) $(1 + i)(2 - 3i)$

$$g(x) = \begin{cases} \frac{k^2-2x}{3-x} & \text{if } -\infty < x < 2 \\ 12 & \text{if } x = 2 \\ 4x+k & \text{if } 2 < x < \infty \end{cases}$$

removable d/s at $x=2$

find k



$$\lim_{x \to 2} g(x) = 12 = g(2)$$

$$\lim_{x \to 2^-} g(x) = \frac{k^2 - 2(2)}{3-2} = k^2 - 4$$

$$\lim_{x \to 2^+} g(x) = 4(2) + k = 8 + k$$

$$k^2 - 4 = 8 + k$$

$$k^2 - k - 12 = 0$$

$$(k-4)(k+3) = 0$$

$$k = 4 \quad k = -3$$

Check $g(2) = 12$

if $k = 4$, the common Limit is 12

$k = -3$

Curve $y = \sqrt{10 - x^2}$ $P(1, 3)$ find tangent line

$= (10 - x^2)^{\frac{1}{2}}$

$y' = \frac{1}{2}(10 - x^2)^{-\frac{1}{2}} \cdot (-2x)$

$= \frac{-2x}{2\sqrt{10 - x^2}}$

$y'(1) = \frac{-(1)}{\sqrt{10 - 1^2}}$

$= \frac{-1}{3}$

$y - y_1 = m(x - x_1)$

$y - 3 = -\frac{1}{3}(x - 1)$

$y - 3 = -\frac{1}{3}x + \frac{1}{3}$

$y = -\frac{1}{3}x + \frac{10}{3}$

$\frac{1}{3} + \frac{3}{1}$

$\frac{10}{3}$

$\cos 2x$

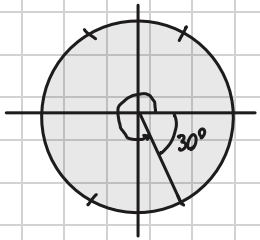
$\cos 2\frac{11\pi}{6}$

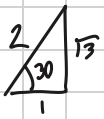
$\cos 2\left(2\pi - \frac{\pi}{6}\right)$

$2\left(1 - \frac{1}{2}\right)$

$= 2\left(\frac{1}{2}\right)$

$= 1$







if $4x - 9y = 0$

$y = \frac{4}{9}x$

$y = \frac{4}{9}\left(\frac{2}{3}\right)$

$= \frac{8}{27}$

$\frac{4}{9} = x^2$

$\sqrt{\frac{4}{9}} = x$

$\frac{2}{3} = x$

$\left(\frac{2}{3}, \frac{8}{27}\right)$

$y = \frac{1}{3}x^3 + c$

$y' = x^2$

$y = \frac{1}{3}x^3 + c$

$\frac{8}{27} = \frac{1}{3}\left(\frac{2}{3}\right)^2 + c$

$\frac{8}{27} - \frac{1}{3}\left(\frac{2}{3}\right)^3 = c$

$\frac{8}{27} - \frac{1}{3} \cdot \frac{8}{27} = c$

$\frac{8}{27} - \frac{8}{81} = c$

$\times 3 \quad \frac{24}{81} - \frac{8}{81} = c$

$\times 3 \quad \frac{16}{81} = c$

$y = x^3 - x - 1 \qquad P(0,-3)$

$y' = 3x^2 - 1$

$m = y'(a) = 3a^2 - 1$

$(a, a^3 - a - 1)$

$y - y_1 = m(x - x_1)$

$y - (a^3 - a - 1) = (3a^2 - 1)(x - a)$

Plug $(0,-3)$

$-3 - (a^3 - a - 1) = (3a^2 - 1)(-a)$

$-2 - a^3 + a = -3a^2 + a$

$-2 + 2a^3 = 0$

$a^3 = 1$

$a = 1$

$m = 3(1)^2 - 1$

$= 2$

$y + 3 = 2(x - 0)$

$y = 2x - 3$

11. Find the slope of the line tangent to the curve $x^3y = 1$ at the point $\left(3, \frac{1}{27}\right)$

$\frac{1}{y} = x^3$

$y = \frac{1}{x^3} = x^{-3}$

$y' = -3x^{-4} = -\frac{3}{x^4}$

$y'(3) = -\frac{3}{3^4} = -\frac{3}{81} = -\frac{1}{27}$

$y - y_1 = m(x - x_1)$

$y - \frac{1}{27} = -\frac{1}{27}(x - 3)$

$y = -\frac{1}{27}x + \frac{4}{27}$

$P(1, 1)$

$x \sin(xy - y^2) = x^2 - 1$

$x^2 - (x \sin(xy - y^2)) - 1 = 0$

$\downarrow \qquad \downarrow$

$2x$

$\sin(xy - y^2) + x \cos(xy - y^2) \cdot (2y)$

$f'(x) = 2x + \sin(xy - y^2) + x \cos(xy - y^2) \cdot (2y)$

$f'(1, 1) = 2(1) + \sin(0) + \cos(0) \cdot (2)$

$= 2 + 0 + 2$

$= 4$

$y - y_1 = m(x - x_1)$

$y - 1 = 4(x - 1)$

$y = 4x - 3$

implicit differentiation

2.9

(for two variables in equation when finding d/dx)

woah

$k^2x^2 - 1 \qquad 2k^2x \qquad 12x^3 - k^2 = 2k^2x$

$\frac{d}{dx}(k^2x^2 - 1) \quad 12x^3 - k^2 \qquad 12 - k^2 = 2k^2$

$12 = 3k^2$

$4 = k^2$

$k = \pm 2$

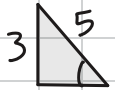
$2k \cdot \frac{dk}{dx} \cdot x^2 + k^2 \cdot 2x$

Implicit differentials

$$y^2x^2 - 1 = 0$$

$\sin\theta = \frac{3}{5}$ $\frac{opp}{hyp}$

$\sin\theta + \cos\theta$



$x^2 + 3^2 = 5^2$

$x^2 = 5^2 - 3^2$

$= \sqrt{5^2 - 3^2}$

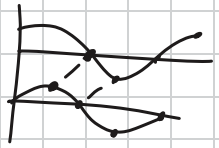
$= \sqrt{25 - 9}$

$= \sqrt{16}$

$= 4$

S A
T C

$-\cos\theta = \frac{4}{5}$



$1 + 0$

$1 + \frac{1}{2}$

$0 + (-1)$

$-1 + 0$

$0 + 1$

$-1.000001 - 1 = -2.000001$

$5\left(\frac{2}{5}\right) + 2$

$\frac{10}{5} = 2$

$$\frac{\frac{4}{5}+5}{4} \qquad \frac{4}{5}+\frac{5}{1} \qquad \frac{29}{5}$$

$$\frac{29}{5} \div \frac{4}{1} \qquad \frac{29}{20}$$

$$\frac{x^2}{x+1} - \frac{x^2}{x-1}$$

$$\frac{x^2(x-1)-(x^2(x+1))}{(x+1)(x-1)} \qquad x^3-x^2-x^3-x^2$$

$$\frac{-2x^2}{x^2-1}$$

$$\frac{-2}{1-\frac{x^2}{x^2}} \qquad \frac{-2}{1-1}$$

$$\frac{-2}{1} \quad -2$$

$$\frac{x^2-1}{|x^2-1|}$$

$$3-m = 1-m3$$
$$-m = 1-m3-3$$
$$-m = -3m-2$$
$$m = 3m-2$$
$$m-3m = -2$$
$$-2m = -2$$
$$m = 1$$

$$y' = 3x^2-3 \qquad y(1) = 1^3-3(1)$$
$$1-3$$
$$=-2$$
$$0 = 3x^2-3 \qquad y(-1) = -1-(-3)$$
$$= 3(x^2-1) \qquad =2$$
$$3(x-1)(x+1)$$
$$x = 1, -1$$

$y' = 2x(4-x^2) + x^2(-2x)$

$0 = 2x(4-x^2) + x^2(-2x)$

$= -2x^3 - 2x^3 + 8x$

$= -4x^3 + 8x$

$= -4x(x^2 - 2)$

$x = 0,$ $\quad x^2 - 2 = 0$

$x^2 = 2$

$x = \pm\sqrt{2}$

$y = 2(4-2)$

$= 2(2)$

$= 4$

$y = 0$

$-\frac{3}{2} - \frac{1}{1} \quad \frac{-3-2}{2} = \frac{-5}{2}$

$y = (1-2t^2)^{-\frac{3}{2}}$

$y' = -\frac{3}{2}(1-2t^2)^{-\frac{5}{2}} \cdot (-4t)$

$\sin(\cos(\tan x))$

$\cos(\cos(\tan x)) \cdot -\sin(\tan x) \cdot \sec^2 x$

$y = \frac{1}{\cos x}$

$y' = \frac{\sin x}{\cos^2 x} \qquad (\cos x)^2 \qquad 2(\cos x) \cdot \sin x$

$y'' = \frac{\cos x \cdot \cos^2 x - (\sin x \cdot 2(\cos x) \cdot -\sin x)}{\cos^4 x}$

$$\frac{(x-6)^{+16}-7}{x-6} = 9$$

$f(x) = x$

$$x^2 - 9x - 7 = (x-6)(x-9)$$

$$6^2 - 6a - 7 = (6-6)(6-9)$$

$$36 - 7 = 6a$$

$$29 = 6a$$

$$\frac{29}{6} = a$$

$$f(x) = x^2 - \frac{29}{6}x - 7$$

$$f(6) = 6^2 - \frac{29}{6}(6) - 7$$

$$\frac{\cos^3 x - (-\sin^2 x \cdot 2\cos x)}{\cos^4 x}$$

$$\frac{\cos^3 x + \sin^2 x \cdot 2\cos x}{\cos^4 x}$$

$$\frac{\cos x(\cos^2 x + 2\sin^2 x)}{\cos^4 x}$$

$$\frac{\cos^2 + \sin^2 x + \sin^2 x}{\cos^3 x}$$

$$\frac{1 + \sin^2 x}{\cos^3 x}$$

$$29 - \frac{29}{6} \times \frac{6}{1}$$

$$29 -$$

$$\lim_{x \to 6} = \frac{f(x) - 7}{x - 6} = 9$$

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

$a = 6 \quad f(a) = 7$

$$f(6) = 7$$

$$\lim_{x \to 6} f(x) = f(6) = 7$$

$$g'(3) = 4$$

$$g' = \frac{4}{3}x$$

$$g = \frac{2}{3}x^2 + C$$

$$g(3) = \frac{2}{3}(9) + C$$

$$= 6 + C$$

could be anything

$$\lim_{x \to \infty} \left(\sqrt{x^2+x} - x \right)$$

$$\left(\sqrt{x^2+x} - x \right) \times \frac{\sqrt{x^2+x} + x}{\sqrt{x^2+x} + x}$$

$$\frac{x}{\left(\sqrt{x^2+x}\right)+x}$$

$$(x+\sqrt{x})+x$$

$x\sqrt{\frac{1}{x^2}+\frac{1}{x}}$

$x\left(\frac{x}{x}+\frac{x^{\frac{1}{2}}}{x^1}+\frac{x}{x}\right)$

$x\left(1+\frac{1}{x^{\frac{1}{2}}}+1\right)$

$$= \frac{1}{\sqrt{1+\frac{1}{x}^{0}}+1}$$

$$= \frac{1}{2}$$

$$\frac{6}{3-i} \times \frac{3+i}{3+i}$$

$$= \frac{18+6i}{9-i^2}$$

$$= \frac{18+6i}{10}$$

$$= \frac{9}{5} + \frac{3}{5}i$$

$$\frac{2\left(\frac{2}{5}\right)+5}{2+2}$$

$$\frac{4}{5} + \frac{5}{1}$$

$$\frac{29}{5} \quad \frac{29}{20}$$

$$cx^2 - 3 = cx + 2$$

$$c(2)^2 - 3 = c(2) + 2$$

$$4c = 2c + 5$$

$$2c = 5$$

$$c = \frac{5}{2}$$

$$y = 3x^2 - 3$$

$$0 = 3(x^2 - 1)$$

$$= 3(x-1)(x+1)$$

$$x = 1, -1$$

$$y(1) = 1 - 3 + 9 = 7$$

$$y(-1) = -1 + 3 + 9 =$$

$$f(x) = x\tan(x) + C$$

$$y = x^2(4-x^2)$$

$$y' = 2x(4-x^2) + x^2(-2x)$$

$$0 = 8x - 2x^3 - 2x^3$$

$$= -4x^3 + 8x$$

$$= -2x(x^2 - 4) \qquad x^2 - 2 = 0$$

$$x^2 = 2$$

$$x = 0 \qquad x = \sqrt{2}$$

$$y = x^2(4-x^2)$$

$y(0) = 0$

$y(\sqrt{3}) = 2(4-2) = 4$

$$\sec^2(\cos(x^2)) \cdot -\sin(x^2) \cdot 2x$$

$$-2x\sec^2(\cos(x^2))\sin(x^2)$$

$$\lim_{x \to 6} \frac{f(x) - 7}{x - 6} = 6$$

$$\frac{f(6) - 7}{(6) - 6} = 6$$

$$\frac{f(x) - 7}{0} = 6$$

$$f(x) - 7 = 0$$

$$f(x) = 7$$

$g'(1) = 7$

$g =$

$x^3 - y^3 = 27$

$3x^2 - 3y^2 \frac{dy}{dx} = 0$

$3x^2 = 3y^2 \frac{dy}{dx}$

$\frac{x^2}{y^2} = \frac{dy}{dx}$

$\sin(xy - y^2) + x\cos(xy - y^2) \cdot (y + x\frac{dy}{dx} - 2y\frac{dy}{dx}) = 2x$

$0 + 1 \cdot 1 \cdot (1 + \frac{dy}{dx} - 2\frac{dy}{dx}) = 2(1)$

$1 - \frac{dy}{dx} = 2$

$\frac{dy}{dx} = -1$

$y - y_1 = m(x - x_1)$

$y - 1 = -1(x - x_1)$

$y = 2 - x$

$$x^3y + xy^2 = 6$$

$$3x^2y + x^3\frac{dy}{dx} + y^2 + x2y\frac{dy}{dx} = 0$$

$$3x^2y + y^2 = -x^3\frac{dy}{dx} - x2y\frac{dy}{dx}$$

$$\frac{3x^2y + y^2}{-x^3 - x2y} = \frac{dy}{dx}$$

$$3x^2y + y^2 = 0$$

$$-3x^2y = y^2$$

$$-3x^2 = y$$

$$-3x^2 = 0$$

$$x^2 = 0$$

$$\lim_{x\to\infty} \sqrt{x^2+5x} - x$$

$$\sqrt{x^2+5x} - x \cdot \frac{\sqrt{x^2+5x} + x}{\sqrt{x^2+5x} + x}$$

$$\frac{x^2 + 5x - x^2}{\sqrt{x^2+5x} + x}$$

$$\frac{5}{1 + \frac{5^0}{x} + 1}$$

$$\frac{5}{2}$$

$$\frac{2-i}{2+i} \cdot \frac{2-i}{2-i}$$

$$\frac{(2-i)(2-i)}{4-i^2} = \frac{(2-i)^2}{5} = \frac{4-4i+i^2}{5} = \frac{3-4i}{5} = \frac{3}{5} - \frac{4}{5}i$$

($i^2 = -1$)

$$\frac{x^2(x-1) - x^2(x+1)}{x^2-1}$$

$$x^3 - x^2 - x^3 - x^2$$

$$\frac{-2x^2}{x^2-1}$$

$$\frac{-2}{1-\frac{1}{x^2}}$$

$$= -2$$

$$x - m = 1 - mx$$

$$-m + 3m = 1 - 3$$

$$-m(1-3) = -2$$

$$-m = \frac{2}{-2}$$

$$m = -1$$

$$\cot\theta = \frac{1}{\tan\theta}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$\csc = \frac{1}{\sin\theta}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

$$\sin^2\theta = \frac{1-\cos(2\theta)}{2}$$

$$\cos^2\theta = \frac{1+\cos(2\theta)}{2}$$

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \frac{1-\cos\theta}{\sin\theta}$$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$1 + \tan^2\theta = \sec^2\theta$$

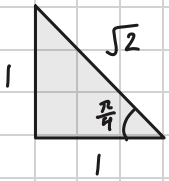
$$1 + \cot^2\theta = \csc^2\theta$$

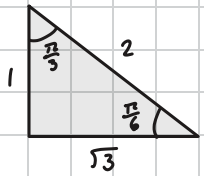
$$\sec^2\theta - 1 = (\sec\theta - 1)(\sec\theta + 1)$$

$$\sin(A+B) = \sin(A)\cos(B) + \cos(A)\sin(B)$$

$$\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$$

sign flips







1. $\sin^2 x + \cos^2 x$

Your answer...

2. $\frac{1-\cos^2 x}{\sin x}$

Your answer...

3. $\frac{\sin x}{\cos x}$

Your answer...

4. $\sec^2 x - \tan^2 x$

Your answer...

5. $\frac{1}{\csc x}$

Your answer...

$$\sin^2 x + \cos^2 x = 1$$

$$\frac{1-\cos^2 x}{\sin x}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\frac{\sin^2 x}{\sin x} = \sin x$$

$$\frac{1-\cos^2 x}{\cos x} = \cos x \tan^2 x$$

$$= \cos x$$

$$\frac{\sin^2 x}{\cos x}$$

$$\cancel{\cos x} \cdot \frac{\sin^2 x}{\cos^{\cancel{2}} x}$$

$$\frac{\sin x}{\cos x} \cdot \sin x$$

$$\frac{\sin^2 x}{\cos x}$$

$$\tan x \sin x = \cos x \tan^2 x$$

$$\sin x \tan x$$

$$\tan x \sin x = \sin x \tan x$$

$$\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$$

$$\frac{\sin x}{1+\cos x} \cdot \frac{1-\cos x}{1-\cos x}$$

$$\frac{\sin(1-\cos x)}{1-\cos^2 x}$$

$$\frac{\sin(1-\cos x)}{\sin^2 x}$$

$$\frac{1-\cos x}{\sin x}$$

$$\lim_{x \to 0} \frac{\tan x}{\sin x}$$

$$\frac{\frac{\sin x}{\cos x}}{\sin x}$$

$$\frac{\sin x}{\cos x} \times \frac{1}{\sin x}$$

$$\frac{1}{\cos x}$$

$$\frac{1}{1} = 1$$

$$\lim_{x \to 0} \frac{\sin(5x)}{\sin(2x)}$$

$$\frac{\sin(5x)}{2\sin x \cos x}$$

$$\frac{5\sin x \cos x}{2\sin x \cos x}$$

$$= \frac{5}{2}$$